

Loan Analysis Dashboard using Power BI

1. Introduction

The Loan Analysis Dashboard project is an end-to-end Business Intelligence solution developed using Power BI, SQL Server, Power BI Service, and DAX. The main objective of this project is to analyze loan data, identify financial risk patterns, monitor loan defaults, and generate meaningful business insights through interactive visualizations.

This project demonstrates real-world data analytics workflow including:

- Data extraction
- Data transformation
- Data modeling
- DAX calculations
- Dashboard creation
- Dataflow implementation
- Incremental refresh configuration
- Scheduled refresh automation
- Report publishing in Power BI Service

The dashboard helps in understanding customer financial behavior, loan trends, default rates, and credit score analysis.

2. Objectives of the Project

The main objectives of this project are:

- To analyze loan distribution across different customer categories
 - To identify loan default patterns
 - To evaluate financial risk metrics
 - To understand customer demographics and financial profiles
 - To perform Year-over-Year (YOY) loan analysis
 - To automate report refresh using Power BI Service
 - To implement incremental refresh for large datasets
 - To create interactive dashboards for business decision-making
-

3. Tools and Technologies Used

The following tools and technologies were used in this project:

Tool/Technology	Purpose
Power BI Desktop	Dashboard creation and visualization
Power BI Service	Report publishing and refresh scheduling
SQL Server	Data storage and management
Power Query Editor	Data cleaning and transformation
DAX	Data analysis and calculations
Dataflow	Centralized cloud-based data management
Gateway	Secure data refresh connectivity
Incremental Refresh	Optimized dataset refresh

4. Dataset Description

The dataset used in this project contains loan-related information of customers. It includes demographic, financial, and loan performance details.

Important Columns in Dataset

- Loan Amount
- Loan Purpose
- Employment Type
- Income
- Age
- Marital Status
- Education Type
- Credit Score Category
- Mortgage Status
- Dependents
- Loan Default Status
- Year

The dataset was imported into SQL Server and later connected to Power BI through Dataflows.

5. Data Import and Dataflow Creation

The project workflow started with importing the dataset into Microsoft SQL Server. After successful import, a Dataflow was created using Power BI Service.

The Dataflow was then connected to Power BI Desktop to ensure centralized data management and easier refresh handling.

Steps Performed

1. Installed and configured SQL Server
 2. Imported dataset into SQL Server
 3. Created Dataflow in Power BI Service
 4. Connected Power BI Desktop to Dataflow
 5. Loaded data into Power BI Desktop
-

6. Data Cleaning and Transformation

Data cleaning and preprocessing were performed using Power Query Editor.

Transformations Performed

- Data type corrections
- Column profiling
- Handling null values
- Renaming columns
- Creating calculated columns
- Data validation checks

These steps improved data quality and ensured accurate dashboard analysis.

7. Dashboard Development

The dashboard was divided into three major pages for better business understanding and analysis.

7.1 Loan Default Overview

This page provides insights into overall loan distribution and default analysis.

Visualizations Included

- Loan Amount by Purpose
- Average Income by Employment Type
- Default Rate by Employment Type
- Average Loan Amount by Age Group

- Default Rate by Year

Key Insights

- Certain employment categories showed higher default rates
 - Loan distribution varied significantly across purposes
 - Default rates changed over time
-

7.2 Applicant Demographics & Financial Profile

This page focuses on customer demographic and financial characteristics.

Visualizations Included

- Median Loan Amount by Credit Score Category
- Donut Chart for Average Loan Amount by Age Group and Marital Status
- Loan Analysis by Credit Categories
- Mortgage and Dependents Analysis
- Loan Analysis by Education Type

Key Insights

- Credit score impacts loan eligibility and loan amount
 - Marital status and age influence financial behavior
 - Education level affects borrowing patterns
-

7.3 Financial Risk Metrics

This page provides advanced financial analysis and trend monitoring.

Visualizations Included

- YOY Loan Amount Change
- YOY Default Loan Change
- YTD Loan Amount Analysis
- Decomposition Tree Analysis

Key Insights

- YOY trends showed fluctuations in loan performance
 - Financial risk patterns were identified across customer groups
 - Decomposition Tree enabled root-cause analysis
-

8. DAX Measures and Calculations

Several advanced DAX measures and calculated columns were implemented in this project.

Important DAX Functions Used

- CALCULATE
- FILTER
- SUMX
- AVERAGEX
- MEDIANX
- DIVIDE
- SWITCH
- ALLEXCEPT
- VALUES
- COUNTROWS

Purpose of DAX

- Perform business calculations
 - Calculate YOY metrics
 - Compute default rates
 - Analyze financial performance
 - Build dynamic visualizations
-

9. Incremental Refresh Implementation

Incremental Refresh was configured to optimize dataset refresh performance.

Benefits

- Faster refresh time
- Reduced system resource usage
- Efficient handling of large datasets

This feature ensures that only newly added or modified data is refreshed instead of refreshing the entire dataset.

10. Scheduled Refresh and Gateway Configuration

A Standard Mode Gateway was installed and configured to enable secure communication between Power BI Service and SQL Server.

Configurations Performed

- Gateway installation
- Gateway connection setup
- Scheduled refresh configuration
- Automatic report refresh setup

This automation ensured that dashboards always display updated information.

11. Report Publishing and Sharing

After successful dashboard development, the report was published to Power BI Service.

Activities Performed

- Published report to workspace
- Configured scheduled refresh
- Shared report access

This enabled cloud-based dashboard accessibility and collaboration.

12. Challenges Faced

During the project development, several challenges were encountered:

- Understanding advanced DAX calculations
- Managing data transformations
- Configuring incremental refresh
- Gateway connectivity issues
- Validating business calculations

These challenges were resolved through testing, validation, and troubleshooting.

13. Conclusion

The Loan Analysis Dashboard project successfully demonstrates an end-to-end Business Intelligence workflow using Power BI and SQL Server.

The project provides meaningful insights into:

- Loan trends
- Financial risk metrics
- Customer demographics

- Default analysis

It also demonstrates practical knowledge of:

- Data transformation
- Advanced DAX
- Power BI Service
- Dataflows
- Incremental Refresh
- Scheduled Refresh

This project improved analytical, visualization, and business intelligence skills and reflects real-world reporting and dashboard development practices.